LESSON 5.6

Represent Three-Dimensional Figures in Two Dimensions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the surface area of a solid by using its net.

WHY THIS MATTERS Your class is designing custom wrapping paper for time capsule boxes. To figure out how much paper you need, you have to find the total surface area of each box. The trick? Unfold the box into a net so you can see every face laid flat! One time capsule box is 6 inches long, 4 inches wide, and 2 inches tall. How much wrapping paper covers the whole box?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Surface area *Spanish: Área de superficie*	The total area of all the flat sides of a solid.	A box 6 × 4 × 2: SA = 2(24) + 2(12) + 2(8) = 88 in² — like wrapping paper needed
Net *Spanish: Plantilla (desarrollo plano)*	A flat shape that folds up into a solid.	unfolded box
Face *Spanish: Cara*	One flat side of a solid shape.	a cube has 6 faces
Two-dimensional *Spanish: Bidimensional*	A flat shape with length and width, but no thickness.	a net on paper
Edge *Spanish: Arista*	The line where two flat sides of a solid meet.	A cube has 12 edges — the lines where two flat faces connect
Pyramid *Spanish: Pirámide*	A solid with one base and triangular sides that meet at a point.	square base + 4 triangles

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Net for a box 5 cm × 3 cm × 2 cm.

1. Bottom and top: 5 cm by 3 cm.
2. Front and back: 5 cm by 2 cm.
3. Two sides: 3 cm by 2 cm.

Answer: 6 rectangles, 3 matching pairs

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now a smaller box: 4 long, 3 wide, 2 tall. How many pairs of faces does it have?

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- This solid has _____ faces shaped like _____ .
- The net needs _____ flat pieces because _____ .
- I attached each _____ to a side of the _____ .
- When the net folds up, the _____ meets the _____ .

Word bank net · face · fold · unfold · base · pyramid · prism · cube · triangle · square · rectangle · two-dimensional · three-dimensional

WATCH OUT!

A common mistake

A common mistake in Surface Area Using Nets is finding the area of only one face from each of the 3 matching pairs and forgetting to double it, instead of counting all 6 faces. For a box that is 9 in × 4 in × 5 in, a student might add just one face from each pair: (9×4) + (9×5) + (4×5) = 36 + 45 + 20 = 101 in², but the net actually has two of each face, so the correct total is 2(36) + 2(45) + 2(20) = 72 + 90 + 40 = 202 in²

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Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each concept about rectangular prisms.

_____ Top & Bottom faces	(A) l × w (two of them)
_____ Front & Back faces	(B) l × h (two of them)
_____ Left & Right faces	(C) w × h (two of them)
_____ Surface area formula	(D) SA = 2lw + 2lh + 2wh
_____ Volume formula	(E) V = l × w × h
_____ Number of faces	(F) 6 faces total

HINT Each face pair uses two of the three dimensions: top/bottom skips height, front/back skips width, left/right skips length.

2 **Write the name of the correct group on each line.**

Put these steps in order for finding the surface area of a 5 × 3 × 2 box.

Groups:

_____ Identify the three face pairs: 5×3, 5×2, 3×2

_____ Find area of each: 15, 10, 6

_____ Multiply each by 2: 30, 20, 12

_____ Add all pairs: 30 + 20 + 12

_____ SA = 62 in²

HINT Read the cards: one identifies face pairs, one finds areas, one doubles, one adds, one states the answer.

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A time capsule box is a rectangular prism with l = 6 in, w = 4 in, h = 2 in. What is its surface area?

(A) 44 in²
(B) 48 in²
(C) 88 in²
(D) 88 in³

HINT Surface area covers all 6 faces of the 6 × 4 × 2 box — it's not the same as the space inside.

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

1. **Name it** *Which measure is asked?*
2. **Write the formula** *Before any numbers.*
3. **Substitute** *Match each letter.*
4. **Unit** *units, sq units, or cubic?*

4 **Circle the letter of the best answer. Show how you know.**

How many faces does a rectangular prism have?

(A) 4
(B) 6
(C) 8
(D) 12

HINT Picture a cereal box in your hands. Turn it: there's a front, a back, two sides, a top, and a bottom.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Surface area is measured in which type of unit?

(A) Cubic units (in³, cm³, ft³)

(B) Square units (in², cm², ft²)

(C) Linear units (in, cm, ft)

(D) No units needed

HINT Surface area measures the flat covering on the outside of a solid — like wrapping paper.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 **Find the mistake. Explain it, then show the correct work.**

Spot the Mistake — A Counted Face Was Skipped

1. **Read** A time capsule box from the net is l = 6 in, w = 4 in, h = 2 in. Find the surface area.
2. **Label the faces** Top/Bottom: 6×4 = 24, Front/Back: 6×2 = 12, Left/Right: 4×2 = 8
3. **Add the faces** SA = 24 + 24 + 12 + 8 + 8 = 76 in² (only one front/back face counted)

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT Count the faces on the net — a box should have 6, not 5.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

The total surface area is 158 in². Which face pair contributes the most area? Why does that make sense when you look at the dimensions?

SENTENCE STARTER The ___ faces contribute the most area (___ in²) because their dimensions ___ × ___ = ___ are the largest. The total SA is ___ + ___ + ___ = ___ in².

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 5.6

Represent Three-Dimensional Figures in Two Dimensions

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

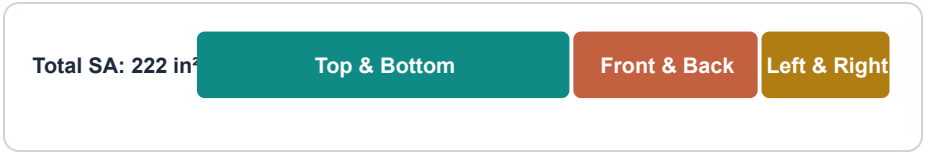
1 **Complete the table. Show how you found each value.**

Find the surface area of each time capsule box using its net.

Box	Length	Width	Height	Surface Area
Box A	10 in	6 in	4 in	
Box B	7 cm	7 cm	3 cm	
Box C	12 ft	5 ft	2 ft	

SHOW YOUR WORK

2 **Use the model to solve. Show your work.**



What is the combined area of the left and right faces?

SHOW YOUR WORK

Answer: _____

3 **Write the name of the correct group on each line.**

Sort these rectangular prisms from LEAST surface area to GREATEST surface area.

Groups:

_____ 2 × 2 × 2

_____ 3 × 3 × 3

_____ 5 × 4 × 3

_____ 6 × 4 × 2

_____ 10 × 1 × 1

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A time capsule box is a cube with edges of 5 cm. What is its surface area?

(A) 30 cm²

(B) 125 cm²

(C) 150 cm²

(D) 150 cm³

Formula _____

Put the numbers in _____

Work it out _____

Answer with its unit _____

PART 3 Explain and correct

reasoning

5 Find the mistake. Explain it, then show the correct work.

Spot the Mistake — One Pair Was Not Doubled

1. **Read** A time capsule box from the net is l = 7 in, w = 5 in, h = 4 in. Find the surface area.
2. **Find each face pair** Top/Bottom: 7×5 = 35, Front/Back: 7×4 = 28, Left/Right: 5×4 = 20
3. **Apply the formula** SA = 2(35) + 2(28) + 20 = 70 + 56 + 20 = 146 in² (left/right pair not doubled)
4. **Answer** Surface Area = 146 in²

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

The total surface area is 158 in². Which face pair contributes the most area? Why does that make sense when you look at the dimensions?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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Challenge · Extension

MASTERY CHECK
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- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Compare volume and surface area. Which measurement is bigger for each prism?

Prism	Volume (cubic units)	Surface Area (square units)	Which is numerically larger?
2 × 2 × 2			
5 × 5 × 5			
10 × 10 × 10			

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Surface Area Error

1. **Dimensions** l = 10 in, w = 4 in, h = 3 in
2. **Formula** SA = 2lw + 2lh + 2wh
3. **Calculate** SA = 2(10×4) + 2(10×3) + (4×3) = 80 + 60 + 12 = 152 in²
4. **Answer** Surface Area = 152 in²

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Two boxes have the same volume of 60 in³. Box A is 5 × 4 × 3. Box B is 10 × 6 × 1. Which box uses less wrapping paper (has less surface area)? Show your calculations and explain why a more cube-like shape might use less material.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

The total surface area is 158 in². Which face pair contributes the most area? Why does that make sense when you look at the dimensions?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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